his day: the nature of the quantum world. Physicists had known for nearly three decades that something was wrong, that a change was desperately needed to understand what was happening in the world of the very small—the world of atoms. But they were working blind. Atoms are simply too small to see through any normal microscope, no matter the magnification. The wavelength of visible light is thousands of times larger than the size of an individual atom. But atoms do give off different colors of light when heated, and different kinds of atoms each have their own distinct spectrum of colors, like a fingerprint. While physicists of the late nineteenth and early twentieth centuries learned to recognize these fingerprints, they didn't understand what kind of internal atomic structure could be producing these spectra. There were hints of mathematical regularity among the spectra, and every so often someone managed to devise a way of understanding part of one of them—most notably, Niels Bohr.

In 1913, inspired by the experimental work of New Zealand–born physicist Ernest Rutherford, Bohr proposed a "planetary" model of the structure of an atom, with a tiny yet massive nucleus surrounded by orbiting electrons. In Bohr's model, the electrons were restricted to a particular set of allowed orbits. Electrons could never be between Bohr's allowed orbits, but they could "jump" from one orbit to another. Each orbit corresponded to a different energy, and, as the electrons jumped, they would emit or absorb light equal to the change in their energy, producing the spectrum seen in the lab. These discontinuous jumps of certain energies were known as *quanta*, from the Latin for "how much"—hence the new science of the atomic world came to be known as "quantum physics."

Bohr's model worked astonishingly well for the simplest kind of atom, hydrogen—so well that Bohr won the Nobel Prize for his idea in 1922. Bohr's model seems simple in retrospect, but that's a result of how profoundly it altered and shaped the idea of an atom. When you hear the word "atom," the cartoon image of electrons orbiting a nucleus that pops into your head is almost entirely due to Bohr. His model was a truly brilliant and original insight into the workings of nature. But it was also incomplete, as Bohr well knew. His model utterly failed to predict the correct spectrum of colors for any other kind of atom, even helium, the next simplest after hydrogen. And even for hydrogen, Bohr's model could only explain so much. It could explain the colors in hydrogen's spectrum but not the